Moustafa Mohammed Abdelwhab

Full-Stack Software Developer

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PROFESSIONAL SUMMARY

Full-stack software developer who builds web applications end to end: React and Next.js front ends, Node.js and NestJS backends with REST APIs, PostgreSQL, MongoDB and Redis data layers, and Docker-based deployment with CI. Experienced in modelling business workflows as server-side state machines, keeping data consistent under concurrent use with transactions and row locking, real-time dashboards over WebSockets, background job queues with retries, payment gateway integration, and role-based access that scopes every user to their own data. Writes unit and end-to-end tests and runs them through CI pipelines.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, SQL, HTML5, CSS3, Python, Java
Frontend	React, Next.js, Vite, Tailwind CSS, TanStack Query & Table, Radix UI, Bootstrap, Framer Motion, responsive design
Backend	Node.js, NestJS, Express, REST API design, Socket.IO, BullMQ job queues, JWT authentication, role-based access, Swagger
Databases	PostgreSQL, MongoDB, MySQL, Redis, InfluxDB, Prisma ORM, Mongoose, Supabase (Auth, Storage, Realtime, Edge Functions)
Testing & DevOps	Jest, Supertest, end-to-end testing, Docker, Docker Compose, GitHub Actions CI, Git, Vercel, AWS
Integrations	Paymob payments and webhooks, Resend transactional email, MQTT

PROJECTS

CargoTrack — Import & Export Logistics Platform

Aug – Oct 2026

NestJS · TypeScript · Prisma · PostgreSQL · Redis · BullMQ · Next.js · React · Tailwind CSS · Jest · Docker · GitHub Actions

- Built a full-cycle order, shipping and payment platform for a China-to-Egypt consolidated freight office: a NestJS REST API that owns every business rule, a BullMQ worker, and a Next.js front end with an office dashboard and a client portal behind one sign-in.
- Modelled the order lifecycle as a server-side state machine (8 states, 3 exception branches) with deposit, QC and balance guards; each transition writes its audit row and notifications in the same transaction, and the UI offers only the moves the server allows.
- Handled concurrency in PostgreSQL: row locks (SELECT … FOR UPDATE) stop a shared container being overbooked by simultaneous users, and bulk operations run all-or-nothing in lock order so overlapping batches cannot deadlock.
- Delivered notifications through a transactional outbox and BullMQ with retries, exponential backoff and a dead-letter queue the office can re-queue from the dashboard; imported legacy spreadsheet data idempotently.
- 228 automated tests (86 unit, 142 end-to-end against real Postgres and Redis) in a GitHub Actions pipeline that also builds the production Docker images; 19-table Prisma schema and 9 recorded architecture decisions.

Source: github.com/Eldalatony/CargoTrack

Glimpse — Event Photo Guestbook | SaaS, live

Jul – Oct 2026

React · Vite · Supabase (PostgreSQL, Auth, Storage, Realtime, Edge Functions) · Paymob · Resend · Vercel

- Built and launched a SaaS for collecting an event's photos: hosts build an event page in a four-step wizard with a live preview and 10 themes, guests upload photos and notes from a QR code with no app or account, and hosts moderate, curate a photo wall and download the album.
- Designed it without a custom API server: business rules live in Postgres as row-level security policies and database functions, so per-event photo and note limits hold for every client, and four Edge Functions handle checkout, webhooks, email and a daily cleanup job.
- Integrated Paymob card checkout with a webhook that verifies each payment and creates the event exactly once; sends host receipts and admin alerts through Resend.
- Consolidated seven theme-specific page implementations into one shared React component driven by theme "skins", so the dashboard preview and the live guest page can never drift apart.

- Shipped a separate admin dashboard on its own subdomain with live updates over Supabase Realtime; covered the data layer with 39 SQL test scenarios across 7 suites.

Live: glimpse-shot.com

SHEMMS — Smart Home Energy Monitoring Platform | Graduation Project — Apr – Aug 2026

Node.js · Express · Socket.IO · Next.js · React · PostgreSQL · InfluxDB · MongoDB · Docker Compose

- Built the web platform for a smart-plug energy monitor: a Node.js/Express backend ingests live device readings, derives power, energy and cost, and streams updates to a Next.js dashboard over Socket.IO, with appliance control, schedules and alerts.
- Designed the REST API (auth, appliances, schedules, alerts, analytics) with JWT authentication, including chart-shaped endpoints that return a whole dashboard view in one round-trip.
- Built a three-database data layer — PostgreSQL for relational records, InfluxDB for time series, MongoDB for aggregates — reconciled by an hourly ETL job that upserts on a unique key so re-runs never double-count.
- Team of six — led the architecture and implemented the backend, data pipeline and deployment of the five-service Docker Compose stack.

Source: github.com/Eldalatony/graduation_project

ADDITIONAL WORK

- **Landing pages & marketing front ends** (freelance, 2023 – present) — responsive, conversion-focused pages with React, Next.js, Tailwind CSS, Framer Motion and GSAP, built for Core Web Vitals and SEO.
- **Competition hosting platform** (2024) — multi-tenant SaaS for player and team registration, bracket scheduling, results tracking and organizer dashboards (React, Node.js, PostgreSQL).

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science — Arab Academy for Science, Technology & Maritime Transport, Cairo — Oct 2022 – 2026

- **Front-End Web Development** — React, Bootstrap, JavaScript, HTML5 & CSS3
- **Cloud Computing** — AWS: EC2, S3, Lambda, IAM, VPC, RDS, CloudWatch

LANGUAGES

Arabic — Native | **English** — Professional working proficiency